JWST/MIRI High-contrast Imaging of η Corvi Exozodi

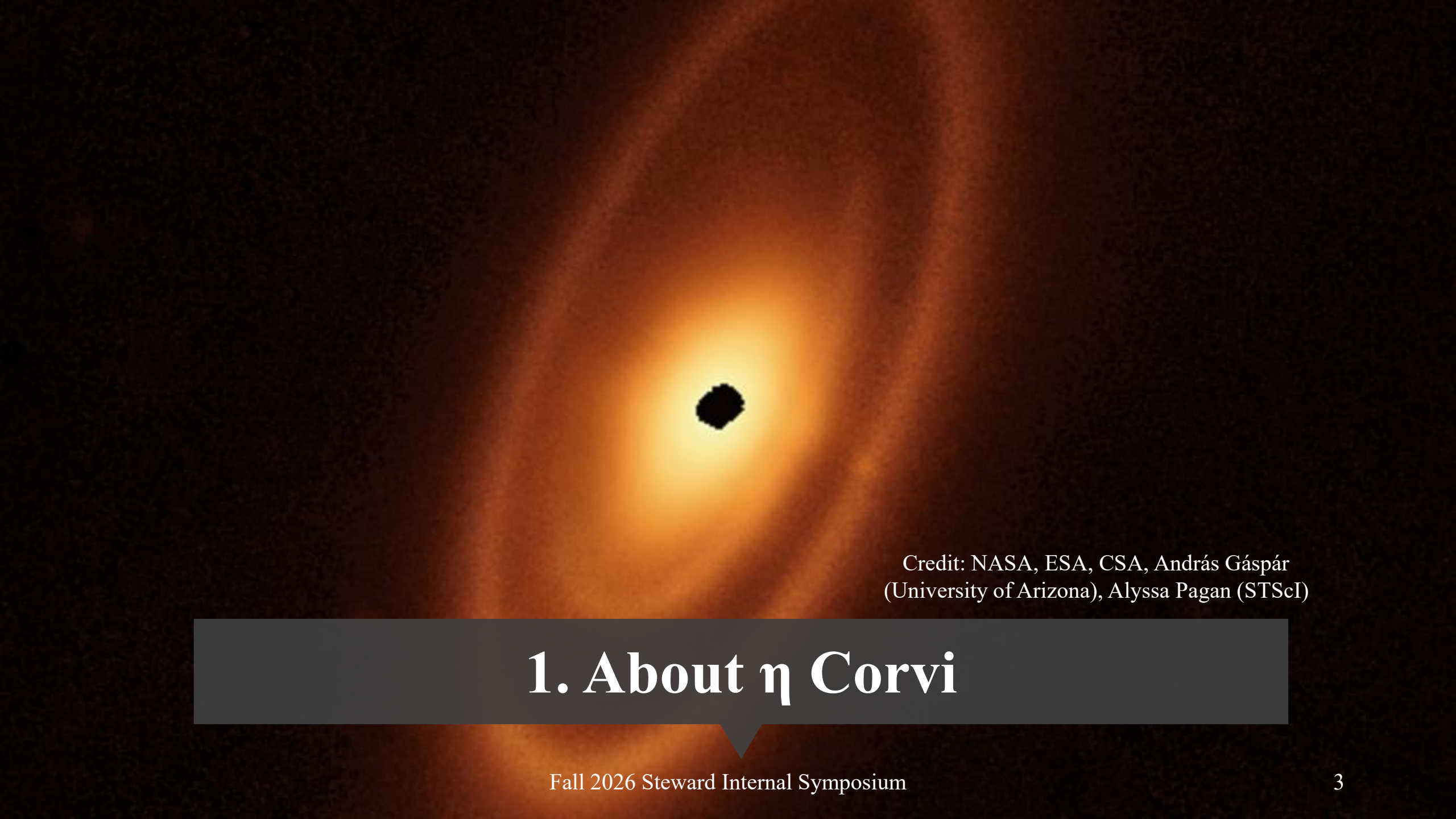
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Agenda

01	About η Corvi
02	Data Acquisition & Analysis
03	Preliminary Results
04	Takeaways

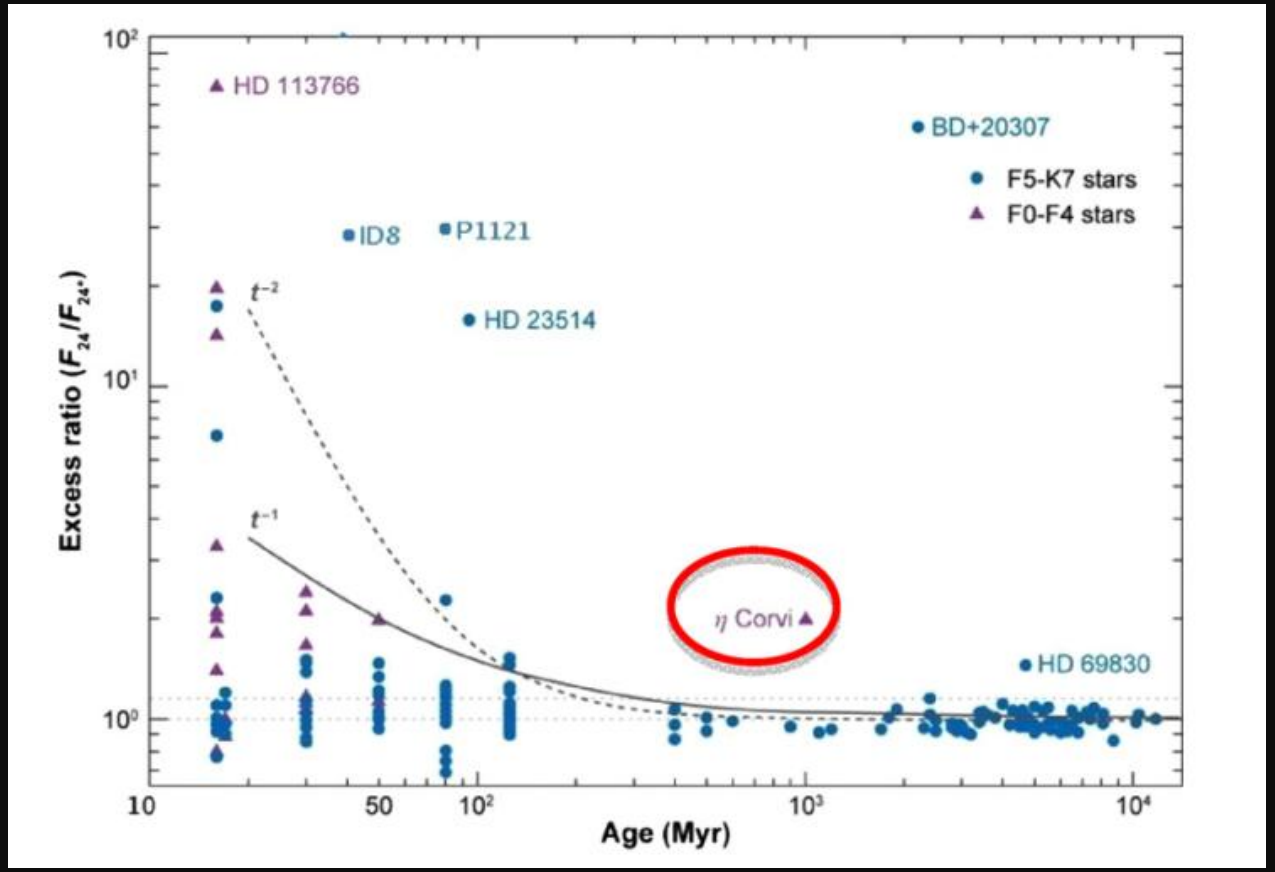


Credit: NASA, ESA, CSA, András Gáspár (University of Arizona), Alyssa Pagan (STScI)

1. About η Corvi

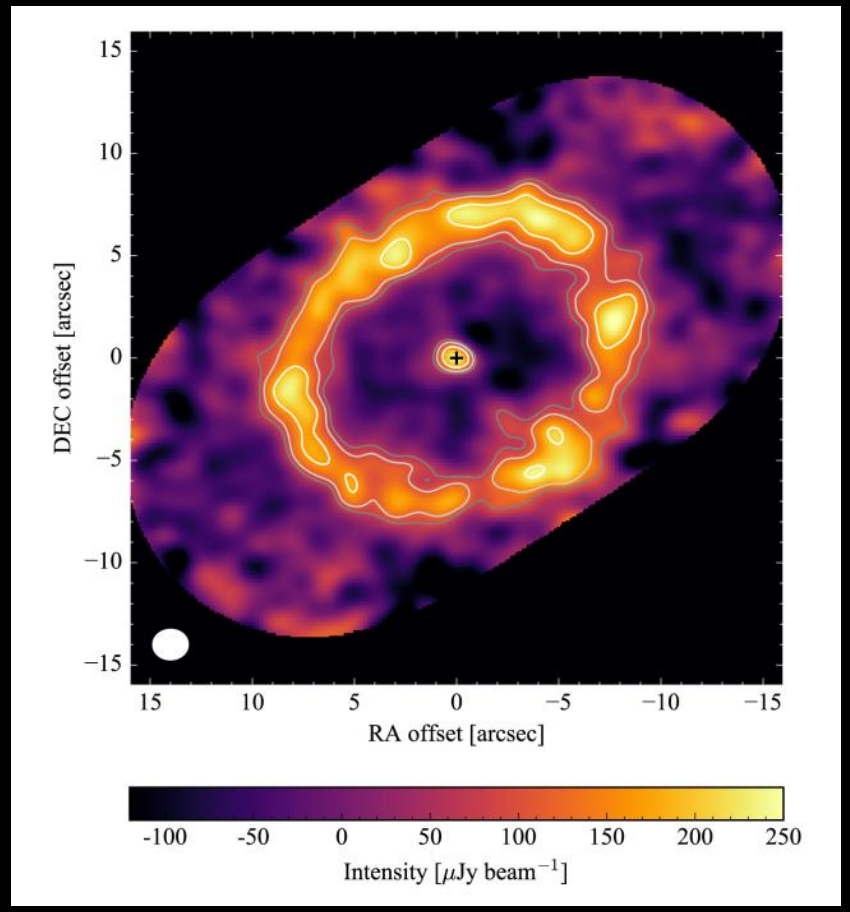
Why η Corvi?

- Stellar Properties (Lebreton et al 2016):
  - Spectral type: F2V
  - $T_{eff} = 6{,}900\ K$
  - $d = 18.02\ pc$
- η Corvi's debris disk is unique with both strong mid- and far-infrared excesses despite its old age (Lebreton+ 2016)
- The disk also has both cold dust and warm dust analog to the Zodiacal dust (Marino+ 2017)
- Previous studies with ALMA and Herschel suggested that the warm dust was delivered into the inner region by Kuiper-like bodies

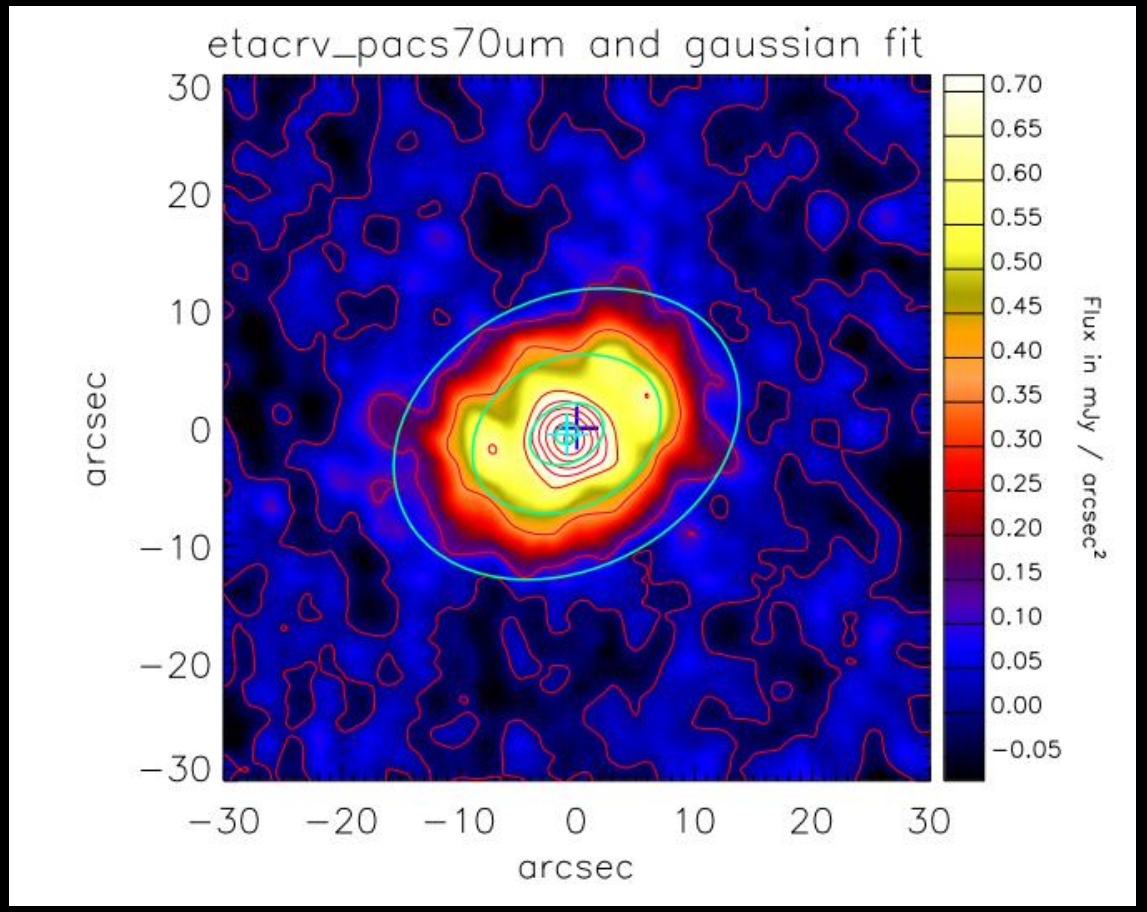


Lisse et al (2012)

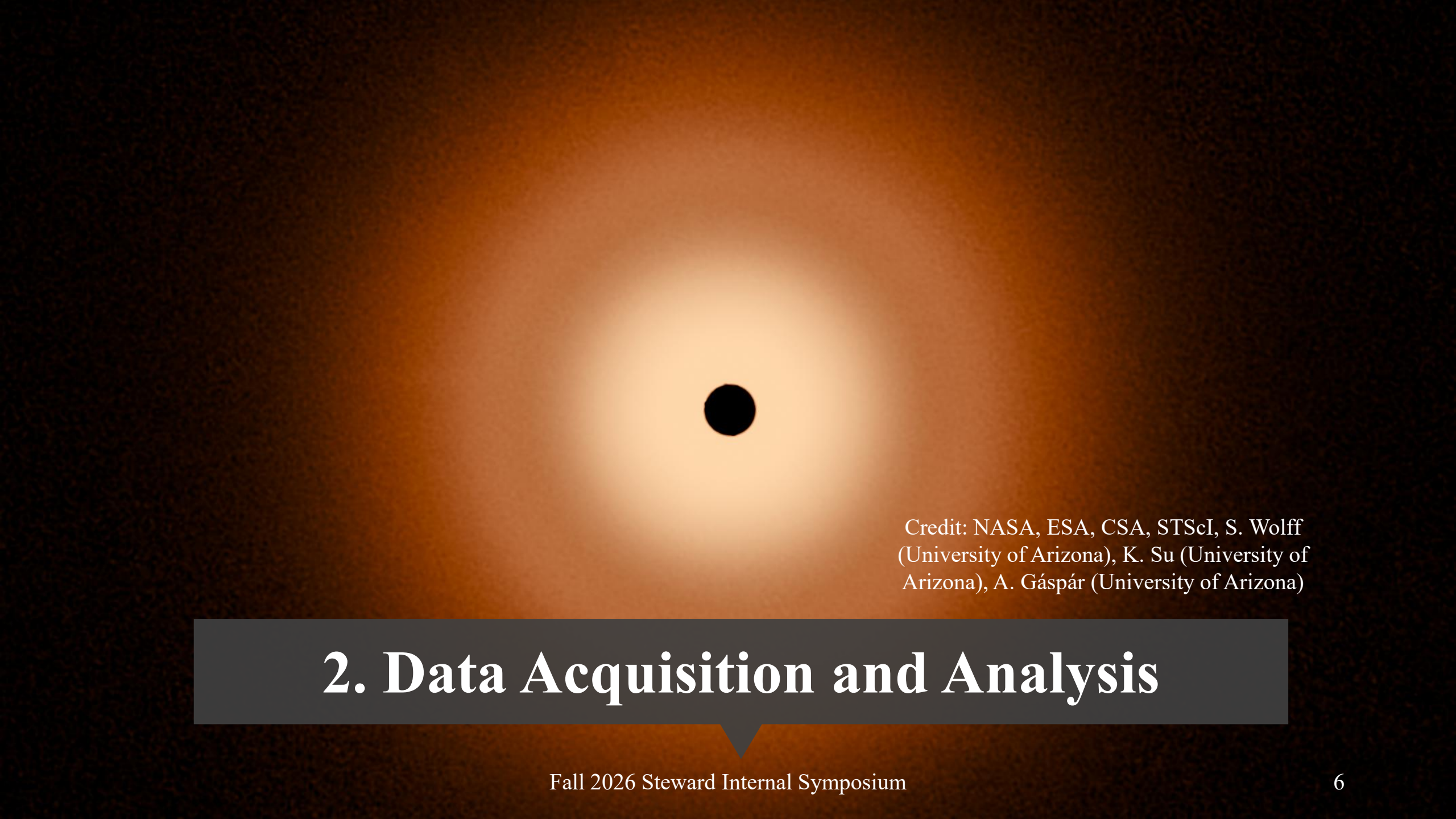
Previous Observations



ALMA (Marino et al 2017)



Herschel (Lebreton et al 2016)

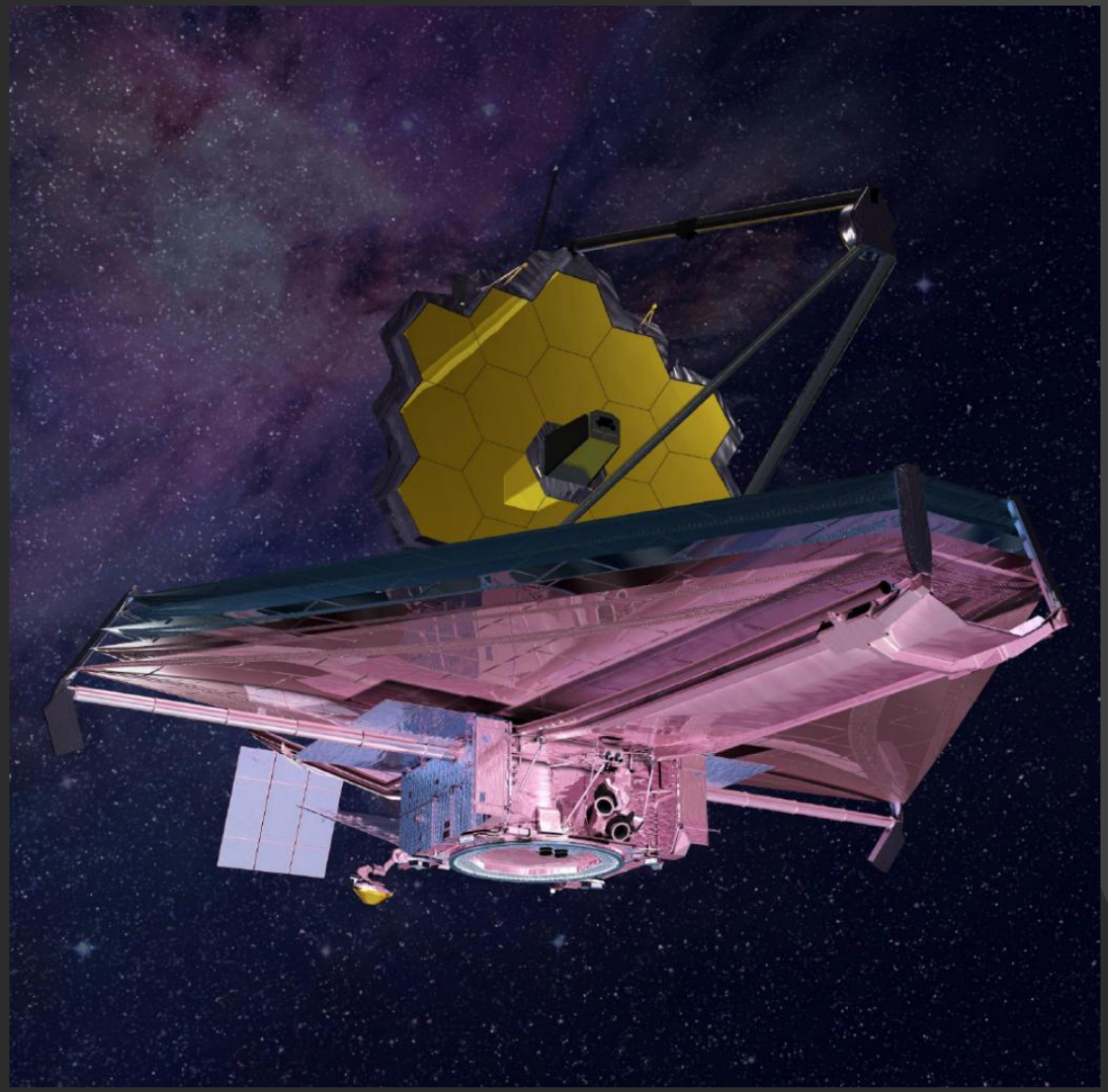


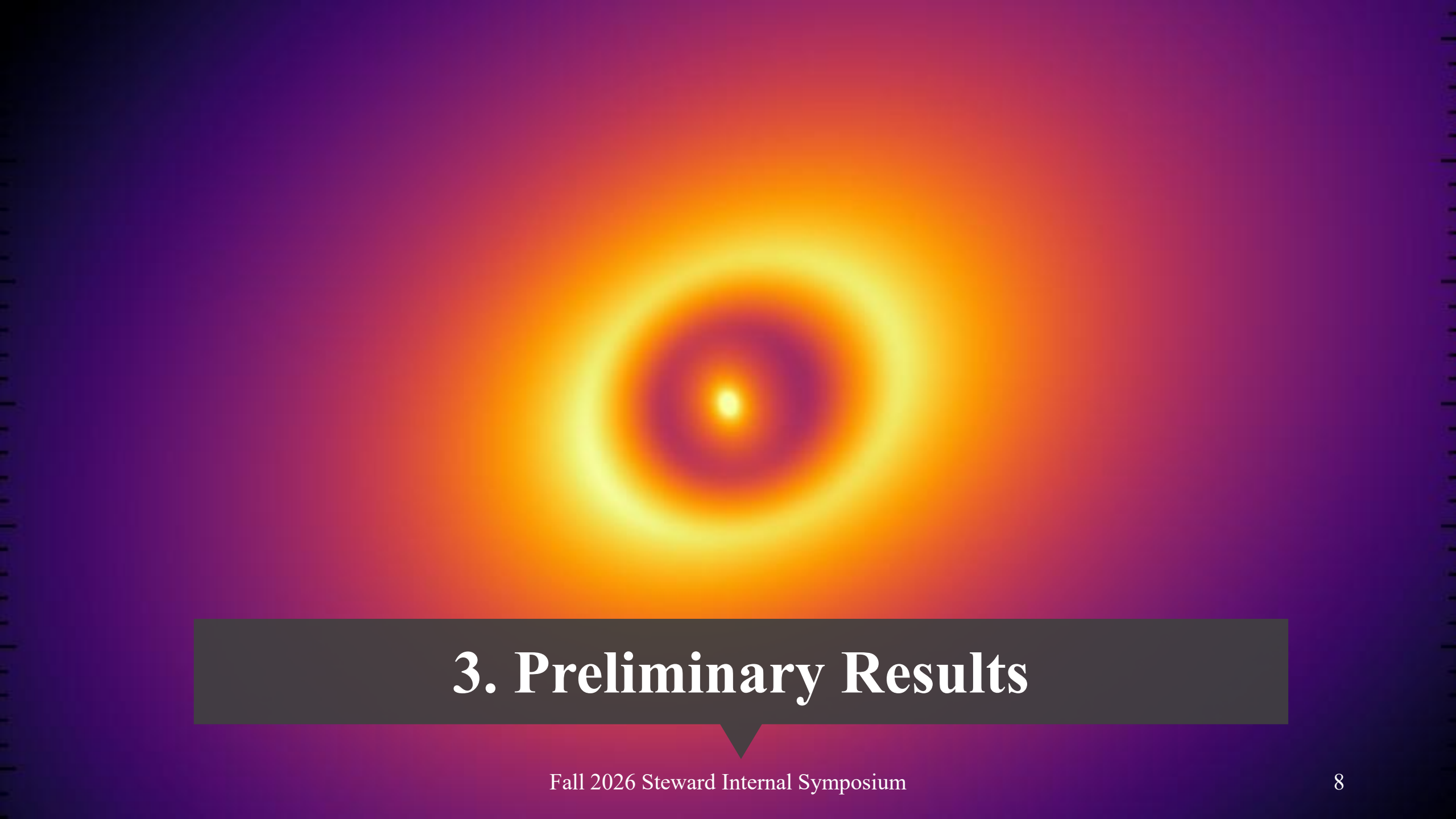
Credit: NASA, ESA, CSA, STScI, S. Wolff (University of Arizona), K. Su (University of Arizona), A. Gáspár (University of Arizona)

2. Data Acquisition and Analysis

Methodology

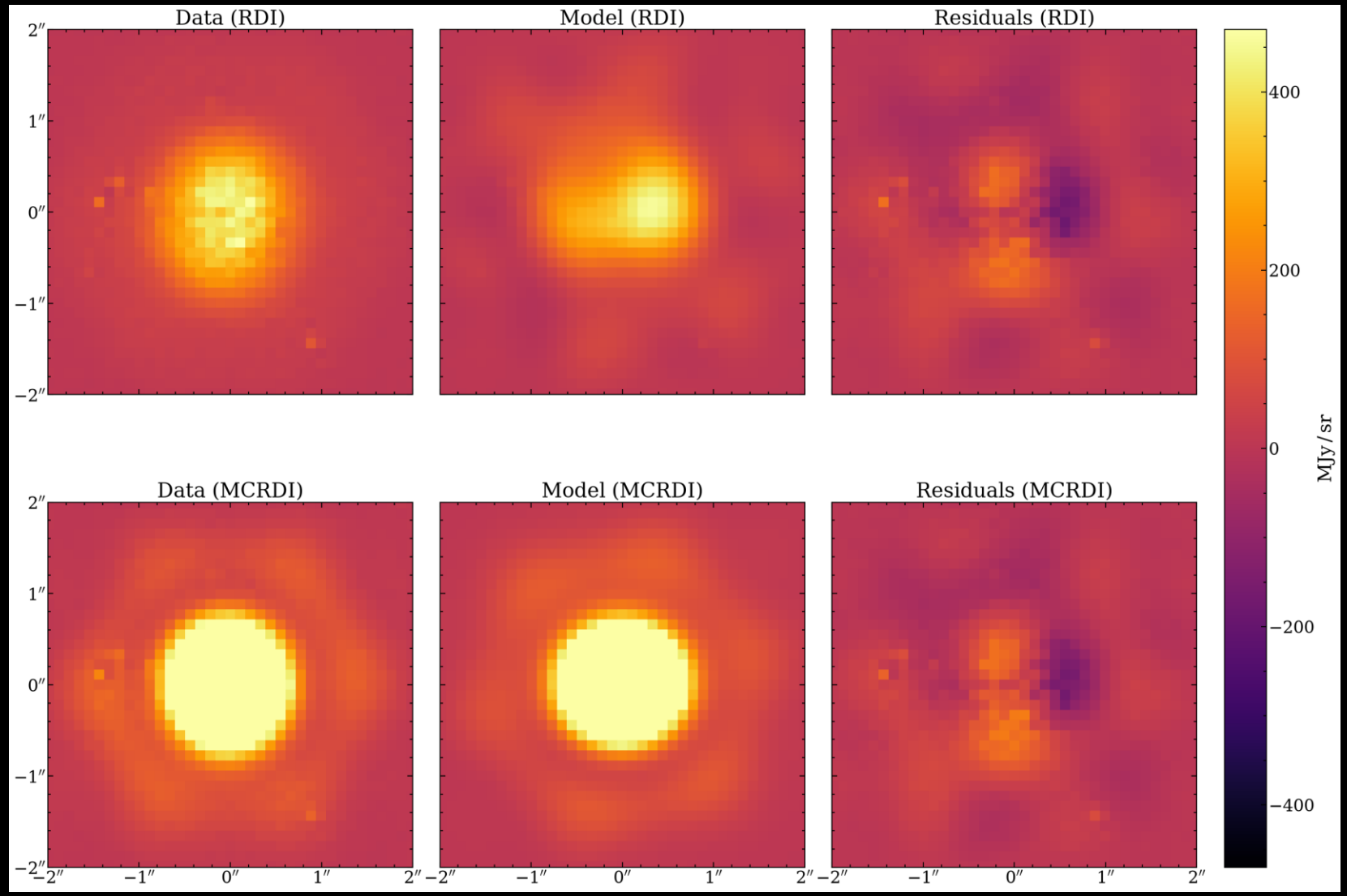
- The dataset is a part of JWST GTO program #4538
- We subtracted the PSF using Reference star Differential Imaging (RDI) to reveal the fainter exozodi
- We forward-modeled the disk image using Levenberg-Marquardt least-square fitting method to determine the best-fit disk parameters.



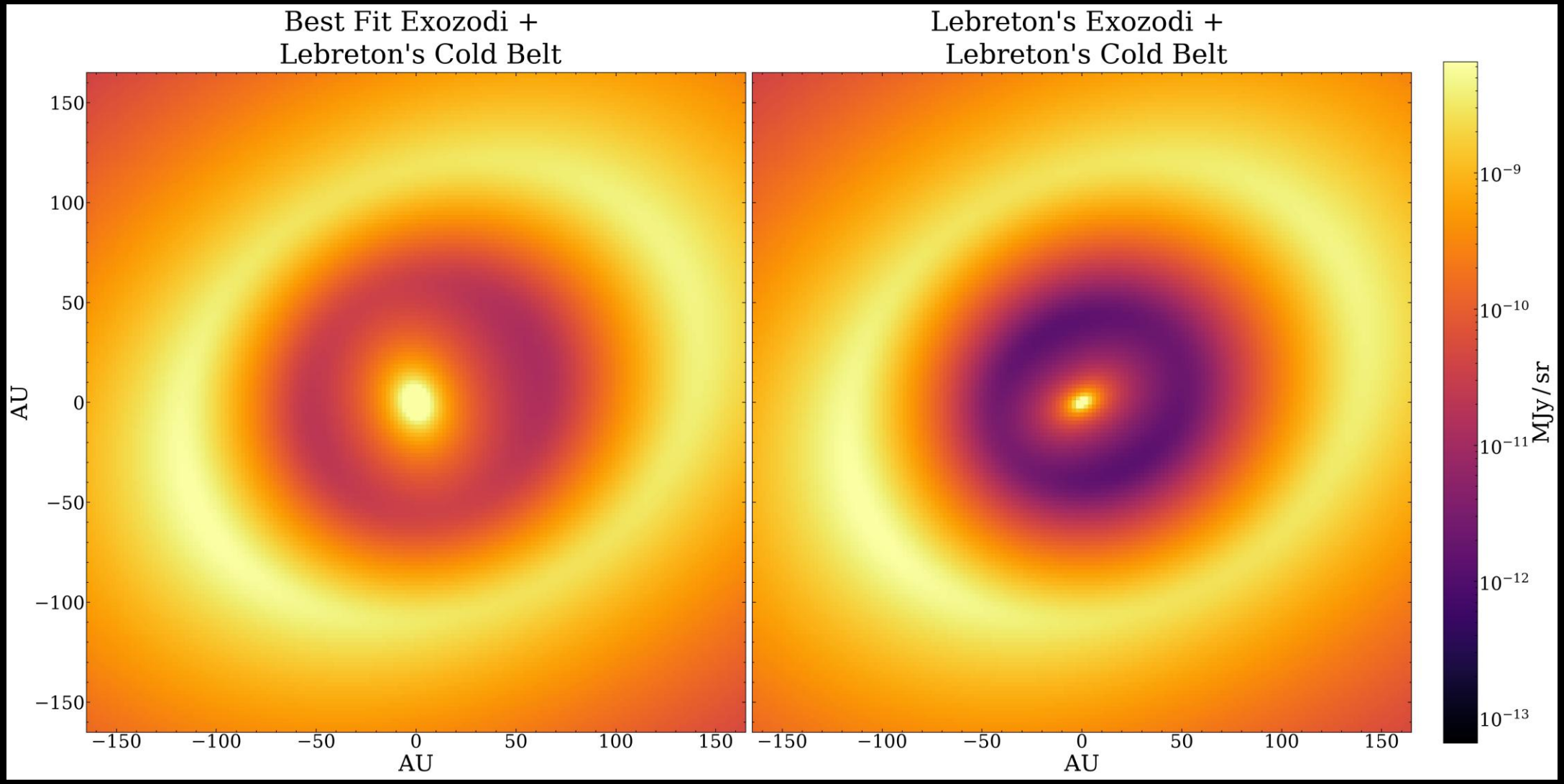


3. Preliminary Results

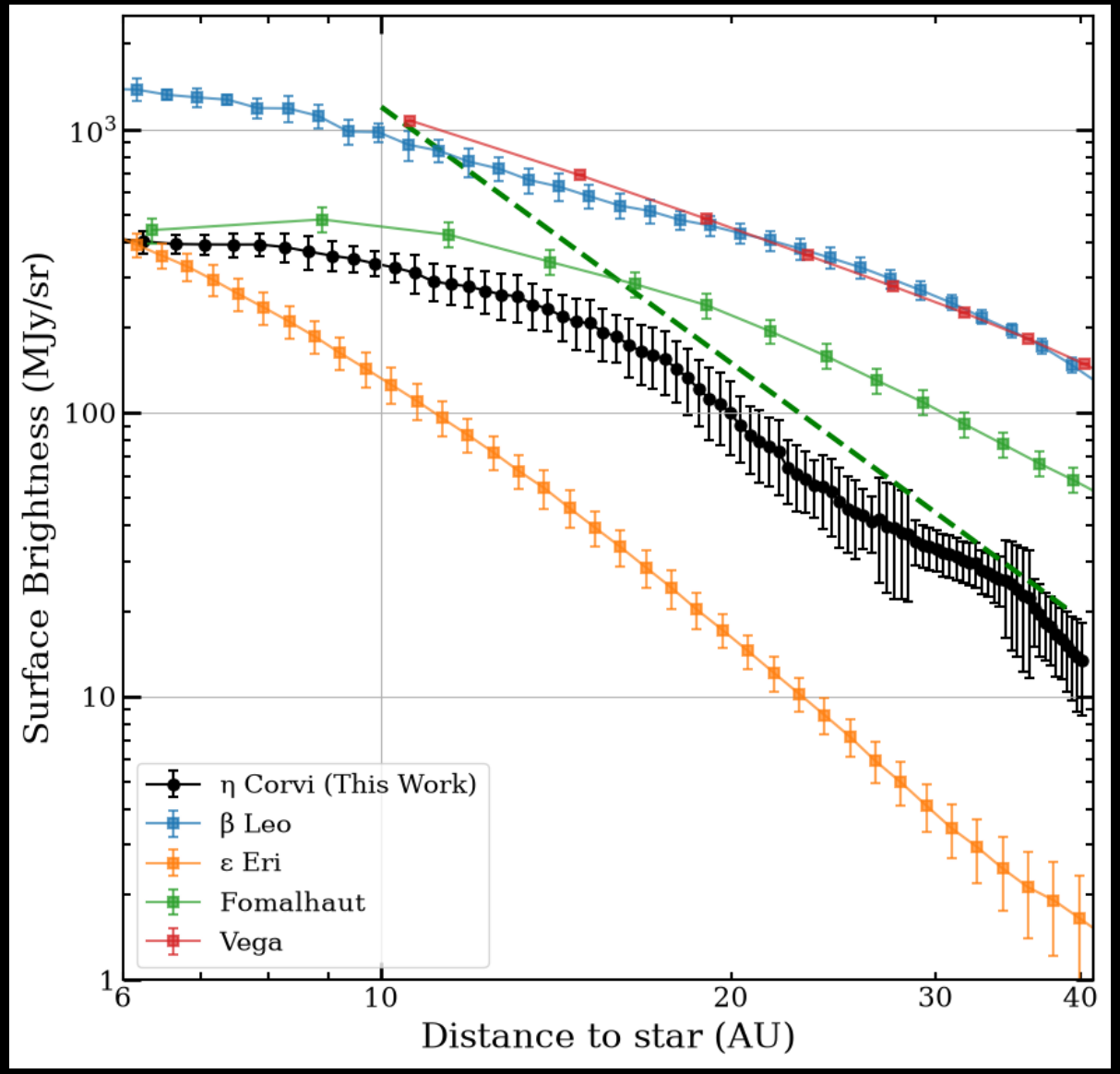
RDI-subtracted Images



Misalignment between Inner and Outer Disks



Evidence for Poynting-Robertson Drag as Exozodi Origin?



References

- Lebreton, J., Beichman, C., Bryden, G., et al. 2016, ApJ, 817, 165
- Lisse, C. M., Wyatt, M. C., Chen, C. H., et al. 2012, ApJ, 747, 93
- Marino, S., Wyatt, M. C., Panic, O. et al. 2017, MNRAS, 465, 2595

Takeaways

- η Corvi is unique as a debris disk host with an outer belt well resolved by ALMA but with evidence for gas, which is usually only seen in younger systems.
- We observed the system using JWST/MIRI F2550W and used RDI subtraction to reveal & forward-model the exozodi.
- The inner exozodi shows clear evidence for misalignment with the previously observed cold disk
- Slope of the radial brightness profile suggests that the warm dust must have been dragged inwards by the host star's radiation under the Poynting-Robertson effect.
- Future implications includes obtaining spectra for the exozodiacal disk with JWST/MIRI MRS to study its chemicalcomposition and gain more insights into its origin.